

LAB EXPERIMENT RECORD

Set 11: Product Category Analysis

Field	Details
Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyse product category sales using a funnel chart, a data table and a pie chart in R, and to build an interactive dashboard that combines them.

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

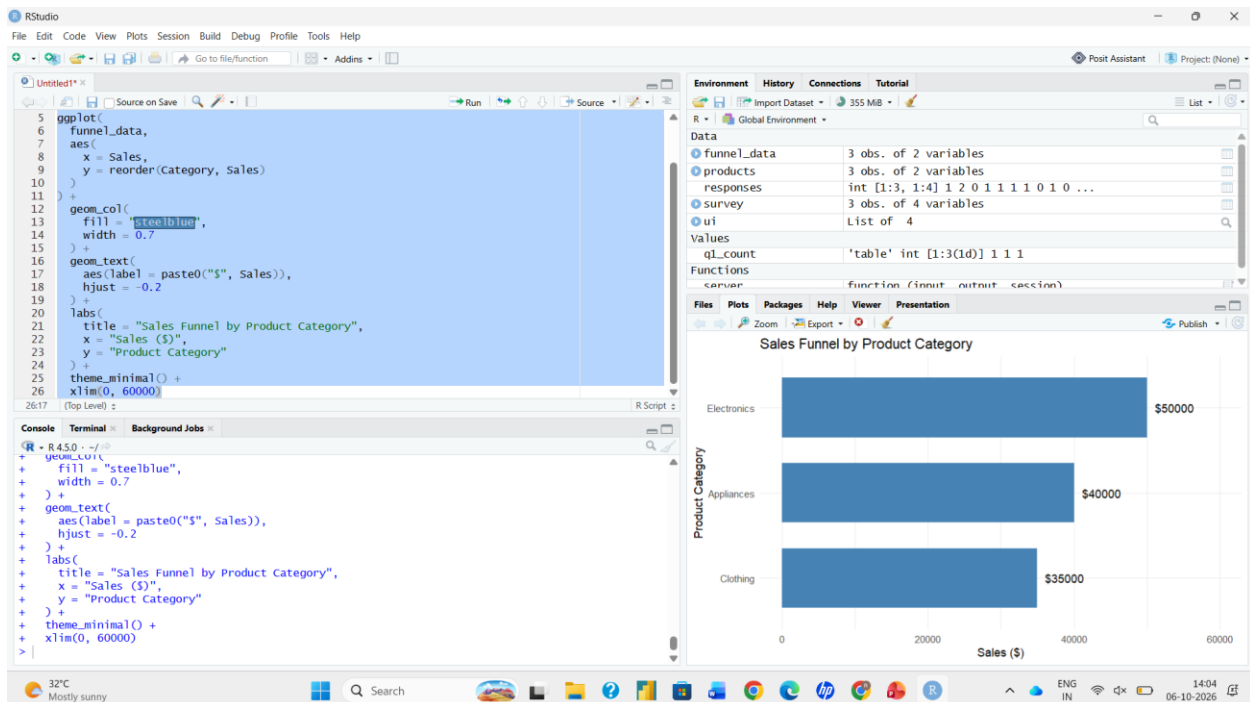
PROGRAM 1: Funnel Chart of Sales

Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
library(plotly)

# Dataset (sorted from highest to lowest sales)
funnel_data <- data.frame(
  Category = c("Electronics", "Appliances", "Clothing"),
  Sales     = c(50000, 40000, 35000)
)

# Funnel Chart
plot_ly(funnel_data,
        type = "funnel",
        y = ~Category,
        x = ~Sales,
        textinfo = "value+percent initial") %>%
  layout(title = "Sales Funnel by Product Category",
         yaxis = list(title = "Stage (Product Category)"),
         xaxis = list(title = "Sales ($)"))
```

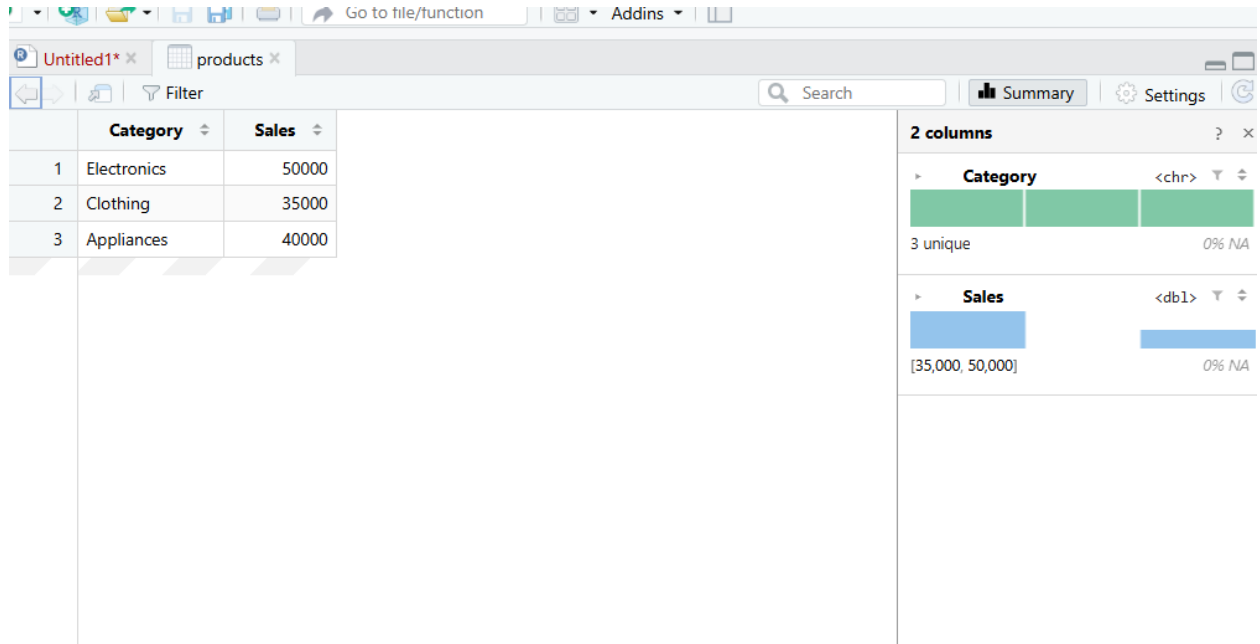


Output 1: Sales Funnel Chart

PROGRAM 2: Product Sales Table

Build a table to display the sales data for each product category. Label the table elements.

```
sales_data <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales     = c(50000, 35000, 40000)  
)  
colnames(sales_data) <- c("Category", "Sales (in $)")  
print(sales_data)  
View(sales_data)
```



The screenshot shows the RStudio interface with a data frame named 'products'. The main editor displays a table with 3 rows and 2 columns. The right sidebar shows the 'Summary' tab with details for the 'Category' and 'Sales' columns.

	Category	Sales
1	Electronics	50000
2	Clothing	35000
3	Appliances	40000

2 columns

Category <chr> 3 unique 0% NA

Sales <dbl> [35,000, 50,000] 0% NA

Output 2: Product Category Sales Table

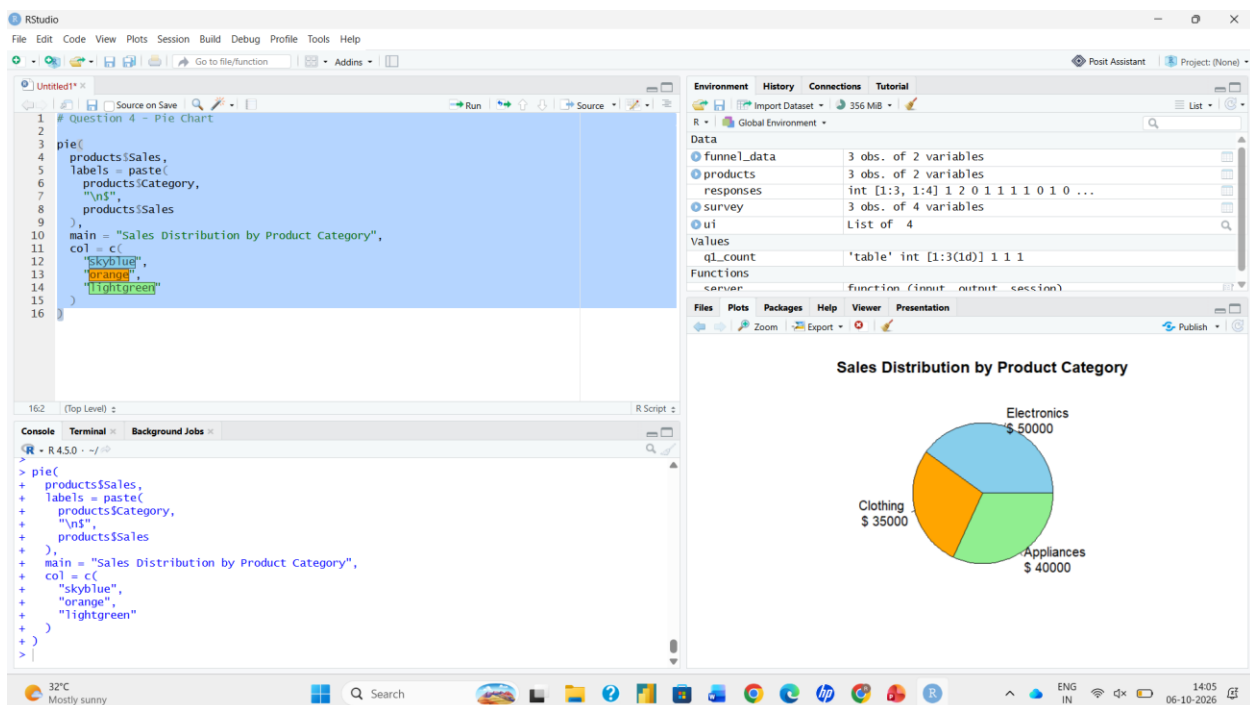
PROGRAM 3: Pie Chart of Sales Distribution

Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
category <- c("Electronics", "Clothing", "Appliances")
sales    <- c(50000, 35000, 40000)

# Percentage labels
pct      <- round(sales / sum(sales) * 100, 1)
labels   <- paste0(category, " (", pct, "%)")

# Pie Chart
pie(sales,
    labels = labels,
    col = c("steelblue", "orange", "seagreen"),
    main = "Distribution of Sales Across Product Categories")
```



Output 3: Sales Distribution Pie Chart

PROGRAM 4: Interactive Shiny Dashboard

Build an interactive dashboard combining the pie chart, funnel chart and the table using the Shiny package.

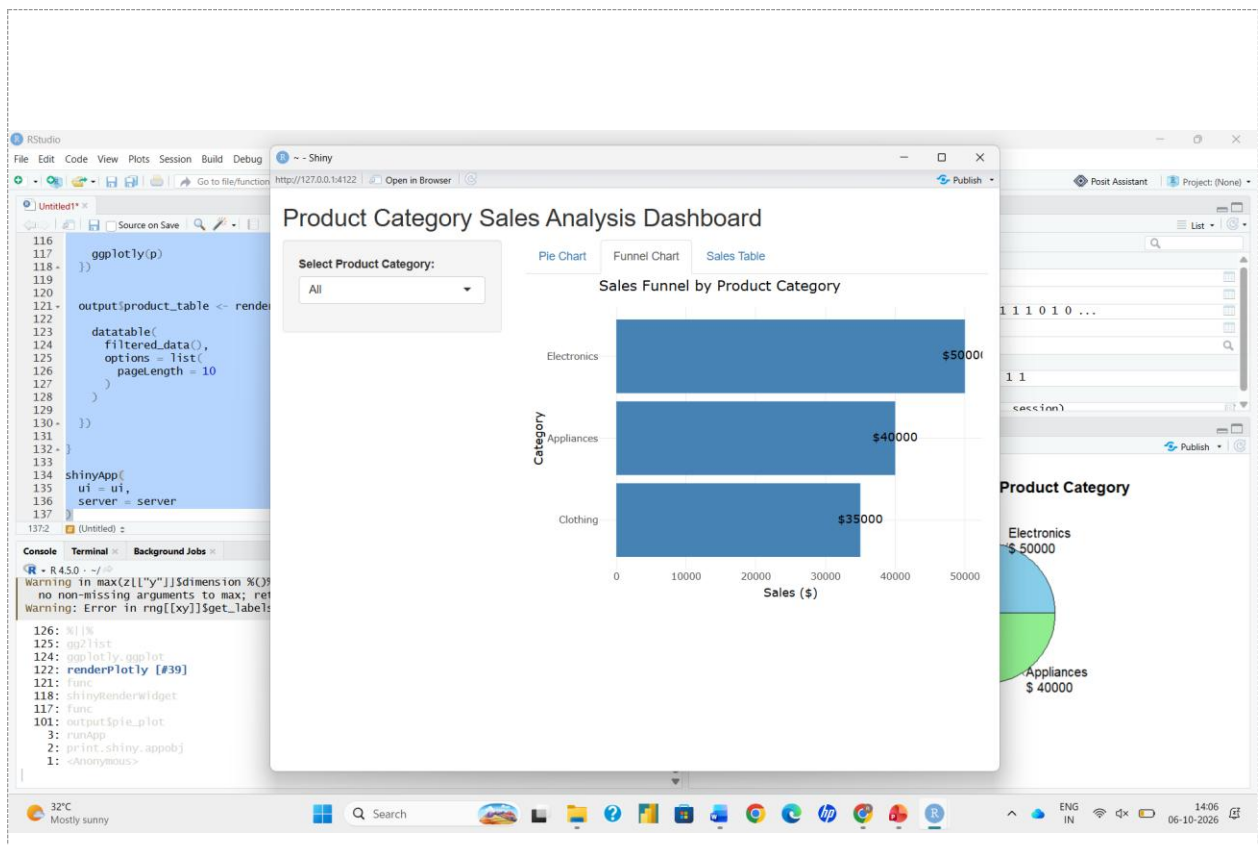
```
library(shiny)
library(plotly)

sales_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales     = c(50000, 35000, 40000)
)

ui <- fluidPage(
  titlePanel("Product Category Dashboard"),
  sidebarLayout(
    sidebarPanel(
      checkboxGroupInput("cat", "Select Categories:",
                        choices = sales_data$Category,
                        selected = sales_data$Category)
    ),
    mainPanel(
      plotOutput("pie"),
      plotlyOutput("funnel"),
      tableOutput("tbl")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({
    d <- sales_data[sales_data$Category %in% input$cat, ]
    d[order(-d$Sales), ]
  })
  output$pie <- renderPlot({
    d <- filtered()
    pct <- round(d$Sales / sum(d$Sales) * 100, 1)
    pie(d$Sales, labels = paste0(d$Category, " (", pct, "%)"),
        col = c("steelblue", "orange", "seagreen")[1:nrow(d)],
        main = "Distribution of Sales Across Product Categories")
  })
  output$funnel <- renderPlotly({
    d <- filtered()
    plot_ly(d, type = "funnel", y = ~Category, x = ~Sales,
            textinfo = "value+percent initial") %>%
      layout(title = "Sales Funnel by Product Category",
             yaxis = list(title = "Stage (Product Category)"),
             xaxis = list(title = "Sales ($)"))
  })
  output$tbl <- renderTable({
    d <- filtered()
    colnames(d) <- c("Category", "Sales (in $)")
    d
  })
}

shinyApp(ui = ui, server = server)
```



Output 4: Interactive Dashboard